ABT RECERTIFICATION EXAMINATION #16

April 23, 1999

PART B

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1990 and 1995. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-847-8720.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend answer sheets.

Good luck!

Key for the following questions:
Choose the single most appropriate answer

1. Which of the following tests can best detect point mutations?
   A) cytogenetics
   B) micronucleus
   C) specific locus
   D) dominant lethal
   E) heritable translocation

2. Which of the following is LEAST likely to detect liver disorders in rodents?
   A) SGPT activity
   B) serum alkaline phosphatase activity
   C) creatinine phosphokinase (CPK)
   D) serum bilirubin
   E) histopathology

3. Which is NOT correct?
   A) fetal hemoglobin is more easily oxidized than that of a one-year old child
   B) *in utero* diethylstilbestrol exposure has been associated with vaginal cancer
   C) isoniazid is likely to cause neurotoxicity in patients with atypical cholinesterase
   D) chloramphenicol is poorly metabolized in the newborn
   E) succinylcholine is a muscle relaxant

4. Allergic responses to chemicals result from damage to target cells that release which of the following?
   A) histamine
   B) slow reacting substance (anaphylactoid srs-a)
   C) serotonin
   D) kinins
   E) all of the above

5. The state in which most chemicals cross the placenta is
   A) in the protein bound form
   B) as a free chemical
   C) as a metabolite
   D) in the ionized form
   E) in the conjugated form

6. Osmolarity of urine is primarily controlled in the
   A) glomerulus
   B) proximal tubule
   C) loop of Henle
   D) distal tubule
   E) collecting duct

7. The principal site of kidney damage by the analgesic mixture of aspirin and phenacetin is which of the following
   A) proximal tubule
   B) distal tubule
   C) loop of Henle
   D) glomerulus
   E) arterioles

8. Which of the following statements is correct?
   A) percutaneous absorption is most efficient when the absorption agent is highly hydrophilic
   B) hexachlorophene can be more hazardous by dermal application than by oral administration
   C) most of the procarcinogens when applied dermally cause tumors at remote sites because the skin does not have an inducible cytochrome p-450 system
   D) percutaneous absorption is not influenced by the nature of the solvent
   E) photoallergic reactions are more common than phototoxic reactions

9. Measurement of which of the following should NOT be used to assess glomerular function?
   A) para-aminohippurate excretion
   B) inulin clearance
   C) blood urea nitrogen
   D) urinary albumin
   E) plasma creatinine

10. Which of the following substances is most likely to cause cirrhosis in the liver?
   A) alpha-naphthylisothiocyanate (ANIT)
   B) folic acid
   C) halothane
   D) ethanol
   E) tannic acid

11. Which of the following agents exerts its primary nephrotoxic effects on the glomerulus?
   A) phenacetin
   B) amphotericin B
   C) cyclosporine
   D) mercuric chloride
   E) potassium bichromate

12. The relative regional permeability of human skin has been described for hydrocortisone. Ranking from most permeable to least permeable is as follows
   A) scrotum, forehead, palm, plantar foot arch
   B) scrotum, plantar foot arch, palm, forehead
   C) scrotum, palm, plantar foot arch, forehead
   D) all sites are relatively the same in permeability
   E) none of the above

Key for the following questions:
Choose the single most appropriate answer

13. Analysis of fructose levels in human seminal fluid can aid in the diagnosis of abnormal functioning of the
   A) epididymus
   B) sertoli cells
   C) leydig cells
   D) seminal vesicles
   E) germinal epithelium

14. Which of the following is correct with regard to eye irritants?
   A) surfactants cause damage to protein principally through the lowering of surface tension
   B) prognosis for recovery immediately after exposure to mustard gas is always apparent
   C) the pH of acidic and basic solutions is the only factor for the type of lesion produced
   D) pain and dulling of the cornea after exposure to neutral solvents are both associated with lipid solubility
   E) lacrimators inhibit sulfhydryl enzymes but have only transient effects

15. An FDA segment I study is designed to detect effects
   A) caused by accumulated toxicity
   B) on parturition, lactation, neonatal development
   C) on embryonic development and organogenesis
   D) on mating, fertility, implantation, gestation, parturition, lactation, neonatal development
   E) on function and behavior in offspring

16. The phthalate acid ester, di(2-ethylhexyl)phthalate is
   A) a hepatic carcinogen in rats and mice
   B) a non-genotoxic carcinogen, (i.e. not known to have a direct effect on DNA)
   C) known to induce peroxisome proliferation which has been postulated to result in the formation of reactive oxygen species
   D) is a testicular toxicant in rodents
   E) all of the above

17. Glutathione tissue levels in the liver would be expected to markedly affect the hepatotoxicity of
   A) acetaminophen
   B) bromobenzene
   C) carbon tetrachloride
   D) halothane
   E) A and B

18. Which of the following structures is among the first to develop during gestation in the rat?
   A) brain
   B) aorta
   C) urogenital system
   D) pulmonary system
   E) digestive system

19. Drug-induced megaloblastic anemia may be caused by
   A) blockade of the synthesis of pyrimidine nucleotides
   B) reduction of cytochrome b5
   C) glucose-6-phosphate dehydrogenase (G6PD) inhibition
   D) A and B
   E) none of the above

20. All of the following agents alter adrenergic neurohumoral transmission EXCEPT
   A) methyldopa
   B) cocaine
   C) botulinum toxin
   D) phenylephrine
   E) phenoxybenzamine

21. Alopecia is an important clinical diagnostic clue in
   A) thallium toxicity
   B) botulinum toxicity
   C) arsenic toxicity
   D) triorthocresyl phosphate toxicity
   E) methylmercury toxicity

22. Which statement concerning allergic contact dermatitis is true?
   A) occurs as a result of a type III immune response
   B) the elicitation phase is the period when the sensitization develops
   C) the elicitation phase occurs when effector T lymphocytes become activated in response to a hapten-protein complex
   D) strong allergens are most often water soluble with molecular weights greater than 500
   E) the Index of Sensitivity is the relative capacity of an agent to induce sensitization

23. Restrictive pulmonary diseases are those diseases which result in
   A) increased pulmonary compliance
   B) decreased pulmonary compliance
   C) increased resistance to air flow
   D) decreased resistance to air flow
   E) decreased blood flow due to pulmonary vasoconstriction

Key for the following questions:
Choose the single most appropriate answer

24. A nonsense mutation can result in which of the following?
    A) a fully functional gene product
    B) the gain or loss of one or more base pairs in a gene
    C) a nonfunctional gene product because of premature termination of protein synthesis
    D) a fully functional gene with altered substrate specificity
    E) none of the above

25. Particles having a diameter of 5 um or larger when inhaled are largely deposited in the
    A) tracheobronchial region
    B) nasopharyngeal region
    C) bronchioles
    D) alveoli
    E) none of the above

26. The interaction of alkylating agents with DNA can produce which of the following adducts?
    A) O_6-alkylguanine
    B) C_8-alkylguanine
    C) N_7-alkylguanine
    D) all of the above
    E) A and C

27. The use of dantrolene in malignant hyperthermia is based upon its ability to
    A) increase blood flow to muscles, thus increasing the rate at which heat is dissipated
    B) produce an increase in the respiratory rate which reduces the temperature of blood in the lungs and hence the body
    C) block release of calcium within muscle cells hence reducing muscle tone and heat production
    D) increase the rate of ATP synthesis by myocytes which results in muscle tone and heat production
    E) none of the above

28. Which of the following most accurately reflects the hierarchy of sensitivity to toxic agents from the most sensitive to the most resistant?
    A) Clara cells>Type II alveolar cells>Type I alveolar cells
    B) Type II alveolar cells>mucous cells>ciliated cells
    C) Type I alveolar cells>Clara cells>Type II alveolar cells
    D) mucous cells>ciliated cells>Type II alveolar cells
    E) ciliated cells>mucous cells>Type I alveolar cells

29. Thymic atrophy is a hallmark of toxicity for which of the following classes of chemicals?
    A) halogenated aromatic hydrocarbons
    B) polybrominated biphenyls
    C) dibenzodioxins
    D) polycyclic aromatic hydrocarbons
    E) all of the above

30. People with atopy have
    A) diminished threshold for irritation and defective cell-mediated immunity
    B) diminished threshold for irritation and enhanced cell-mediated immunity
    C) increased threshold for irritation and enhanced cell-mediated immunity
    D) increased threshold for irritation and defective cell mediated immunity
    E) increased threshold for irritation and normal cell mediated immunity

31. Which of the following genotoxicity assays is based on a mutation at the thymidine kinase locus?
    A) Ames *Salmonella* assay
    B) mouse lymphoma assay
    C) Chinese hamster ovary assay
    D) mouse micronucleus test
    E) unscheduled DNA synthesis assay

32. The cell type that is primarily responsible for production of collagen in hepatic fibrosis is the
    A) Kupffer cell
    B) hepatocyte
    C) Ito cell
    D) monocyte, following its differentiation into a macrophage
    E) Sertoli cell

Key for the following questions:
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

33. Which of the following is/are true integrative processes of the hypothalamo-pituitary-gonadal axis?

A
1) FSH and LH are glycoproteins synthesized and released from a subpopulation of the basophilic gondadotropic cells of the pituitary gland
2) luteinizing hormone-releasing hormone (LH-RH) acts on gonadotropic cells, thereby stimulating the release of FSH and LH
3) LH-RH and follicle-stimulating hormone-releasing hormone (FSH-RH) appear to be the same substance
4) gonadotropins are not important in the female

34. Which area(s) in the central nervous system are thought to have incomplete protection from blood-borne toxicants?

B
1) choroid plexus
2) frontal cortex
3) area postrema
4) cerebellum

35. The theoretical time for a test compound gas to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the t99. In addition to the equation constant what measurements are needed to calculate the t99?

B
1) chamber volume
2) chamber target concentration of the test compound
3) chamber airflow rate
4) molecular weight of the test compound

36. Which of the following tests are routinely used to measure the effect of immunotoxicants on B-cell activity?

B
1) plaque-forming cell response to sheep red blood cells
2) complement depletion
3) immunoglobulin quantitation
4) delayed hypersensitivity reactions

37. Which of the following suggest(s) that DNA is a critical target in carcinogenicity?

A
1) many carcinogens are also mutagens
2) most, if not all, cancers display chromosomal abnormalities
3) many carcinogens can be metabolized to electrophiles that react covalently with DNA
4) nuclear DNA levels are markedly decreased in cancerous tissues

38. Which of the following compounds produce(s) cholestasis?

A
1) amitriptylene
2) haloperidol
3) chlorpromazine
4) aflatoxin

39. The rate of dermal absorption of toxicants may by facilitated by

E
1) increasing the toxicant's lipophilicity (for example, 2,4-D acid change to 2,4-D ester)
2) increasing skin blood flow
3) choice of solvent/vehicle
4) increasing skin water content

40. Which of the following statements concerning the kidney is/are true?

A
1) phenobarbital and salicylates will be cleared more rapidly from the plasma after IV administration of sodium bicarbonate
2) amphetamines will be cleared more rapidly from the plasma after IV administration of ammonium chloride
3) the mechanism for secretion of organic compounds is localized in the proximal tubule
4) because of high blood flow to the medulla, it is very sensitive to toxic insult

41. Which of the following statements is/are true?

A
1) the blood-brain barrier displays a developmental process of formation during which the selectivity of the barrier becomes greater
2) the blood-brain barrier is formed primarily by a specialized layer of glial cells surrounding the blood vasculature
3) tight junctions are present between adjacent vascular endothelial cells
4) small, charged molecules tend to penetrate the blood-brain barrier readily

Key for the following questions:
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

42. Which of the following is/are true of p53?
1) loss or inactivation of the p53 tumor suppressor gene is a common molecular lesion in human neoplasms
2) introduction of wild type p53 into p53-negative tumor cells inhibits proliferation by implementing a G1 arrest
3) cells from p53 knockout mice fail to arrest in G_1 following gamma irradiation
4) p53 growth-arrest is mediated via p53-dependent transactivation of genes

43. Which of the following is/are true regarding alpha-2u-globulin?
1) production of alpha-2u-globulin is a characteristic of some chemicals that produce renal tumors
2) alpha-2u-globulin is not seen in female rats, mice or most other laboratory animals
3) chemicals that induce alpha-2u-globulin formation do not generally react with DNA and are also negative in short term tests for genotoxicity
4) chemicals that induce the alpha-2u-globulin are generally of greater concern for humans than nephrotoxicants that do not induce this protein

44. Which of the following is/are a correct pairing(s) of a toxicant and its effect?
1) hepatocellular carcinoma - aflatoxin B1.
2) cirrhosis - ethanol
3) cholestasis - contraceptive steroids
4) veno-occlusive disease - pyrrolizidine alkaloids

45. Congenital cleft palate
1) is due to in utero exposure to the causative agent relatively late in the embryonic period of development in animal models
2) is primarily due to initial normal fusion of the palatal shelves and secondary necrosis at the fusion line
3) can be caused by antiepileptic drugs in humans and animal models
4) exhibits relatively constant spontaneous incidences across different human races and across different rat and mouse strains

46. When basing carcinogenic risk assessment on the assumption of no threshold, which of the following statements is/are correct?
1) the shape of the dose-response curve is likely to be linear at low doses
2) the response of the population follows a specific probability distribution function
3) a cancer can develop from a single cell that has undergone a number of transformations
4) the mechanisms causing the control group's cancer are different from those causing the treated group's cancer

47. Blood is carried to the liver by the
1) hepatic vein
2) portal vein
3) lymphatics
4) hepatic artery

48. Which of the following industrial chemicals are high risk hazards for type IV allergic reactions (cell-mediated, delayed hypersensitivity) ?
1) trichlorethylene
2) isocyanates
3) vinyl acetate
4) mercaptobenzothiazole

49. Which of the following are direct acting carcinogens?
1) N-2-fluorenylacetamide
2) triethylenemelamine
3) aflatoxin b1
4) ethylmethane sulfonate

50. In two stage skin carcinogenesis studies, tumor promoting activity has been shown by
1) phenobarbital
2) simple phenols
3) retinoids
4) phorbol esters